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Question 7 (27 marks)

Show your workings clearly.

Question 7a (6 marks)

A distribution network for medical supplies is shown in Figure 2a. The nodes represent locations and numbers on the arcs represent distances between locations.

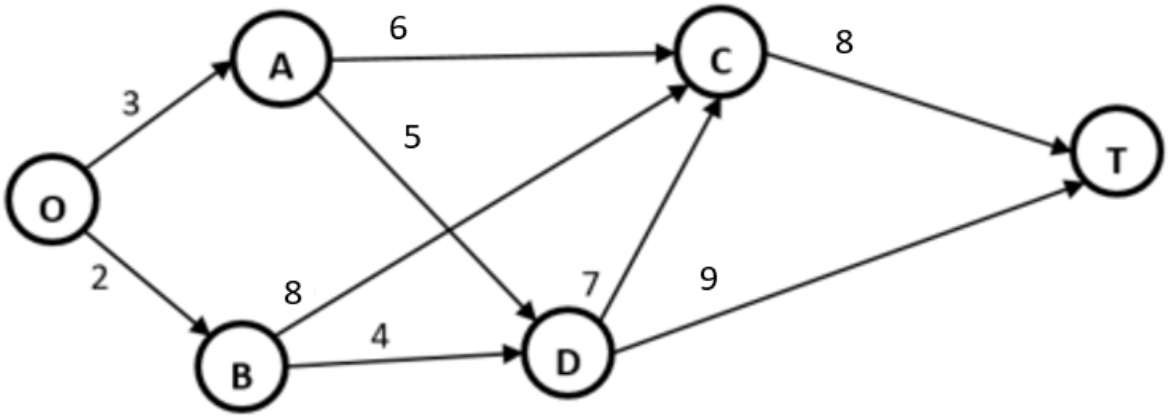

Figure 2a: A Distribution Network

Table 2a shows the partial workings to find the shortest path from node O to each of the other nodes in the network.

Table 2a: Partial Workings for Shortest Path Problem

Iteration	Solution Set	Arcs	Total Distance
Iteration 1	O	O → A O → B	3 2
Iteration 2	O B (2)	O → A B → C B → D	3 10 6
Iteration 3	O B (2) A (3)	B → C B → D A → C A → D	10 6 9 **V**
Iteration 4	…	…	…
…	…	…	…

Question 7a(i) (1 mark)

Name the algorithm used in Table 2a to solve the shortest path problem.

The algorithm used in Table 2a to solve the shortest path problem is **Dijkstra's**

Algorithm_____

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Question 7a(ii) (1 mark)

Calculate the value of **V** in Table 2a.

In Iteration 3, the solution set includes node A. The path to A is O → A with a distance of 3. From A, we explore the arc to D.

- Distance from O to A = 3
- Distance from A to D = 5
- V = 3 + 5 = 8.

Question 7a(iii) (2 marks)

Which node should be added into the solution set at **Iteration 4**? Explain.

(iii) Node added at Iteration 4 Node D should be added to the solution set.

- **Explanation:** Dijkstra's algorithm selects the node with the smallest tentative distance (label) from the currently solved nodes.
  - From the solved set $\{O, A, B\}$, the available paths to unsolved nodes are:
    - From B to D: Distance = (Distance to B) + (Arc B-D) = $2 + 4 = 6$.
    - From A to C: Distance = (Distance to A) + (Arc A-C) = $3 + 6 = 9$.
  - Since 6 is less than 9, Node D is selected next.

Question 7a(iv) (2 marks)

Work out the shortest path from node O to node C and state the total distance.

(iv) Shortest Path from O to C

- **Path:** $O \to A \to C$
- **Total Distance: 9**
  - (Note: The alternative path via D is $O \to B \to D \to C$ = $2 + 4 + 7 = 13$, which is longer).

Question 7b (4 marks)

The network diagram in Figure 2b shows the distribution routes for a medical item from Plant 1 (P1) to Customer 1 (C1). The value on each arc represents the **shipping capacity** along each route. Determine the maximum quantity that can be shipped from P1 to C1, stating clearly the route and shipment quantity along each path.

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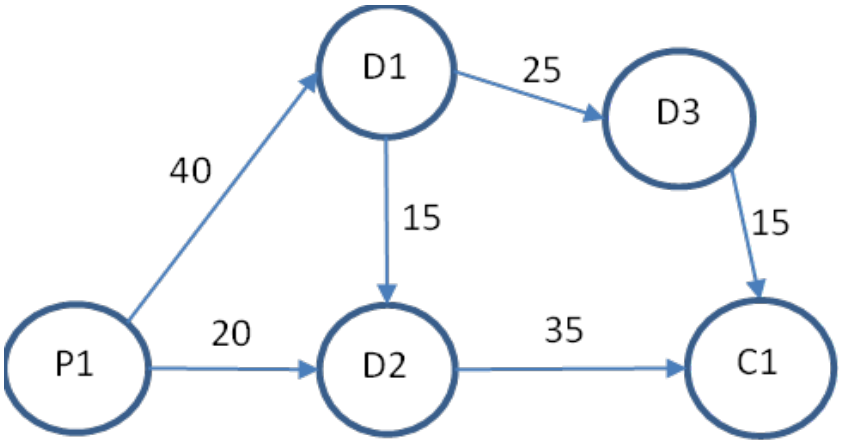

Figure 2b: A Shipping Network Diagram

1. **Route 1:** $P1 \rightarrow D2 \rightarrow C1$
   - Bottleneck capacity: Min(20, 35) = **20**
   - (Arc P1-D2 is now saturated).
2. **Route 2:** $P1 \rightarrow D1 \rightarrow D3 \rightarrow C1$
   - Bottleneck capacity: Min(40, 25, 15) = **15**
   - (Arc D3-C1 is now saturated).
   - Remaining capacity on P1-D1 is $40 - 15 = 25$.
3. **Route 3:** $P1 \rightarrow D1 \rightarrow D2 \rightarrow C1$
   - Remaining capacity P1-D1: 25
   - Capacity D1-D2: 15
   - Remaining capacity D2-C1: $35 - 20 = 15$
   - Bottleneck: Min(25, 15, 15) = **15**

Total Maximum Quantity: $20 + 15 + 15 = \mathbf{50}$.

Question 7c (10 marks)

The manufacturer has two plants (A & B) producing a medical device. The product is shipped to two retailers (D & E) directly or through the transshipment hub at C. The distribution network for shipping the product is shown in Figure 2c where the numbers on the arcs represent unit shipping costs. A Linear Programming (LP) model is formulated to solve the distribution network problem in Figure 2c.

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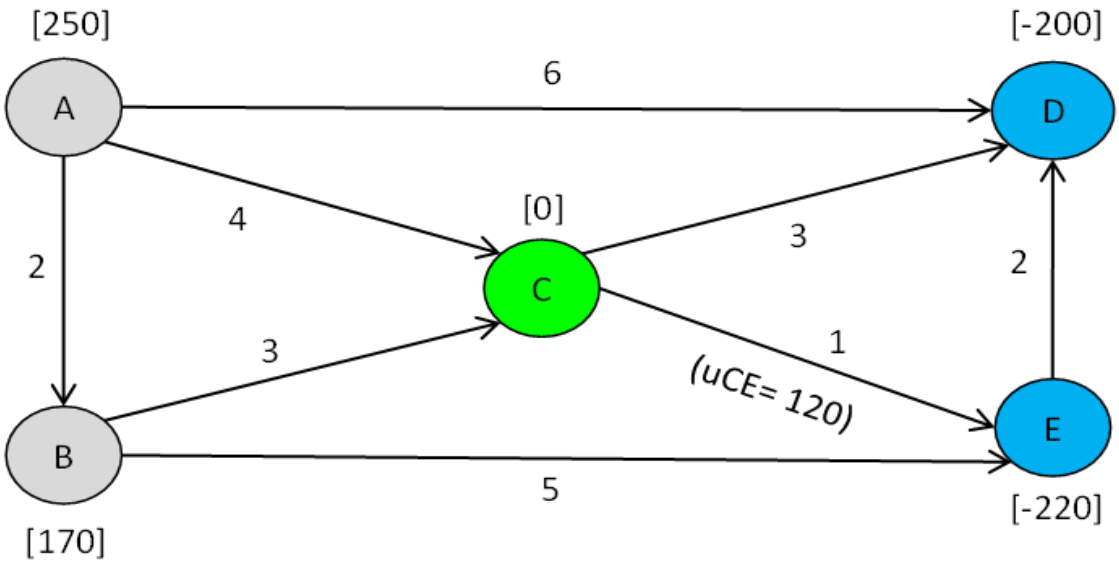

Figure 2c: Medical Device Distribution Network

Question 7c(i) (2 marks)

State in words the **objective** of formulating a distribution plan to ship the product from the plants to the customers.

(i) **Objective** The objective is to **minimize the total shipping cost** (or transportation cost) of distributing the medical devices from the plants to the customers/retailers

Question 7c(ii) (6 marks)

Assume Xij (decision variable) represents the quantity to ship from location i to location j (i, j can be A, B, C, D or E).
Formulate the following constraints:
a. Net flow constraint for **Plant B**. (2 marks)
b. Flow capacity limit from Transshipment Hub C to Retailer E. (2 marks)c. Net flow constraint for **Retailer D if its demand is increased to 250.** (2 marks)

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(ii) Constraints Formulation Based on the network in Figure 2c:

- **a. Net flow constraint for Plant B:** Plant B is a source with supply 170, but it also receives an inflow from A.

$$(\text{Flow Out}) - (\text{Flow In}) = \text{Supply}$$

$$(X_{BC} + X_{BE}) - X_{AB} = 170$$

- **b. Flow capacity limit from Transshipment Hub C to Retailer E:**

$$X_{CE} \leq 120$$

- **c. Net flow constraint for Retailer D (Demand = 250):** Retailer D is a destination (sink) node.

$$\text{Total Flow In} = \text{Demand}$$

$$X_{AD} + X_{CD} + X_{ED} = 250$$

(Note: X_{ED} is included because there is an arc from E to D in Figure 2c).

Question 7c(iii) (2 marks)

List two (2) assumptions when solving the above distribution network using a Linear Programming (LP) model.

- **Linearity:** Costs are directly proportional to the quantity shipped (no volume discounts or fixed costs).
- **Divisibility:** The decision variables (quantities shipped) can be continuous values (fractions are allowed), or
- **Certainty** (parameters like supplies, demands, and costs are known and constant)

Question 7d (7 marks)

In one country, a product is to be distributed from Depot (A) to a number of stores (B, C, D and E). A truck will leave the **Depot (A)**, visit each store once and return to the Depot after delivery. Travel distances between locations are shown in Table 2b.

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Table 2b: Travel Distances Between Locations

From	To (Travel distance in km)				
	A	**B**	**C**	**D**	**E**
A	-	24	18	20	29
B	24	-	22	31	38
C	18	22	-	25	31
D	20	31	25	-	26
E	29	38	31	26	-

Clark and Wright savings method is used to plan a delivery route. Table 2c shows the **Savings** in distance between 2 stores ranked in descending order.

Table 2c: Savings Ranked in Descending Order

Rank	Savings	From	To
1	23	D	E
2	20	B	C
3	16	C	E
4	?	B	E
5	13	B	D
6	13	C	D

Question 7d(i) (2 marks)

Calculate the savings between nodes B and E.

(i) Calculate Savings for B-E The savings formula is $S_{ij} = d_{i0} + d_{0j} - d_{ij}$, where "0" is the depot (A). Using the distances from Table 2b:

- Distance A to B (d_{AB}) = 24
- Distance A to E (d_{AE}) = 29
- Distance B to E (d_{BE}) = 38

$$\text{Savings } (B, E) = 24 + 29 - 38 = \mathbf{15}$$

Question 7d(ii) (2 marks)

Referring to Table 2c, at Rank 1, a sub-tour of A-D-E-A is constructed.
What step should be carried out next at Rank 2?

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(ii) Step at Rank 2 At Rank 1, the sub-tour $A - D - E - A$ was formed. Rank 2 corresponds to the savings between **B and C** (Savings = 20). Since neither B nor C is currently assigned to a route, we **construct a new sub-tour** connecting these nodes: **Sub-tour:** $A - B - C - A$.

Question 7d(iii) (3 marks)

State the complete route based on Clark and Wrights Savings Method and calculate the total travel distance.

(iii) Complete Route and Total Distance We currently have two sub-tours: $(A - D - E - A)$ and $(A - B - C - A)$. We look at the next highest savings to merge them.

- **Rank 3:** Pair C-E (Savings = 16).
- Node C is connected to A in one tour, and Node E is connected to A in the other. We can merge the tours by deleting the A-C and A-E links and adding the C-E link.

Complete Route:

$$A \to B \to C \to E \to D \to A$$

(Or the reverse: $A \to D \to E \to C \to B \to A$)

Total Distance Calculation:

- $A \to B = 24$
- $B \to C = 22$
- $C \to E = 31$
- $E \to D = 26$
- $D \to A = 20$

Total Distance = $24 + 22 + 31 + 26 + 20 = \mathbf{123}$ km.